

HiKey in AOSP - Update

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Overview
Downloading and Building
Establishing a Build Environment
Downloading the Source
Preparing to Build
Compiling with Jack
Selecting Devices
Running Builds
Building Kernels
Known Issues
Developing
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Community

HiKey boards

Google supports HiKey, a certified 96Board, as an Android reference board. AOSP provides kernel source and board support for HiKey to enable developers to easily create and debug new and existing peripheral drivers, do kernel development, and perform other tasks with fewer OEM encumbrances.

HiKey boards are available in 1GB RAM and 2GB RAM configurations from Lenovator:

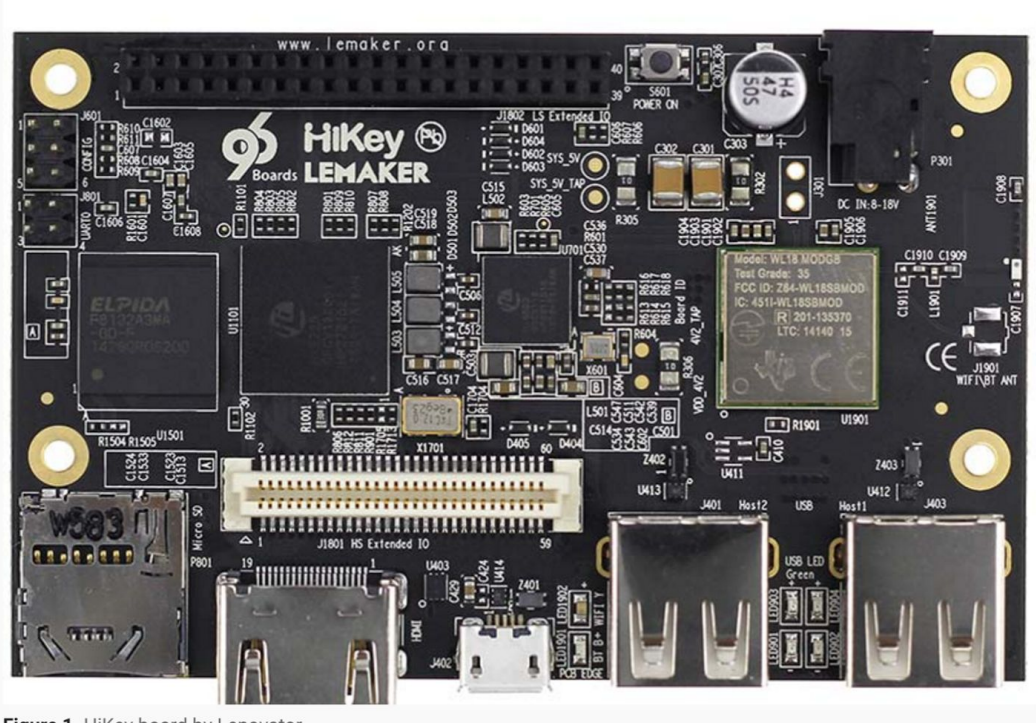


Figure 1. HiKey board by Lenovator

Continuing Collaboration

Working closely with folks at Google.
Submitting changes directly to AOSP Gerrit.

New Features Added Since Announcement

- Updated to Nougat
- v4.4-stable based kernel
- (Staying current w/ -stable)
- Suspend/Resume support
- Interactive cpufreq gov
- powerHAL integration
- HDMI & USB Audio
- Display panel mezzanine
- Bootloader source integration
- Abootimg support in UEFI
- Unique serial no setup
- “fastboot boot <kernel>”
- USB Tethering
- Overlay_Manager
- FIQ_debugger*
- And more...

* FIQ bouncing from secure mode not yet implemented, so really this is an irq_debugger

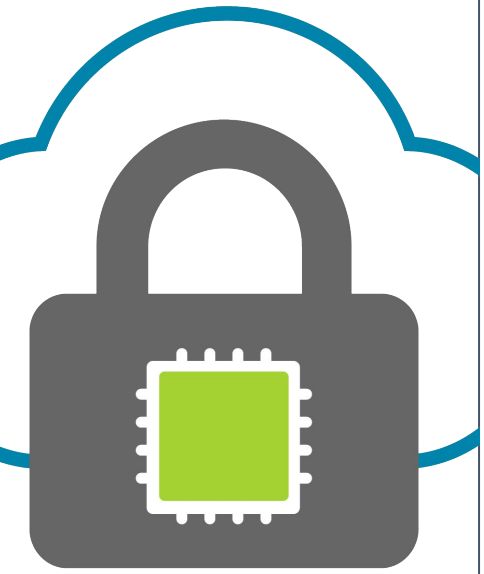


Energy Aware Scheduler Integration

Working through and documenting how to integrate EAS code (core, schedtune, schedfreq gov) in with AOSP & PowerHAL

Working with other Linaro teams to benchmark and measure power and performance impact

Collaboration between Linaro, ARM and Google got EAS merged in android-4.4 and integrated into HiKey!



OPTEE

.org

OP-TEE/Trusty Integration

Working to support both, with co-existance via build time options

OP-TEE kernel driver merged into HiKey AOSP kernel

OP-TEE bootloader integration in progress.

Trusty integration also being actively worked on.

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Overlay Manager

How to handle and support various hardware configurations in AOSP

Driver which switches between DeviceTree overlay objects in the DeviceTree source based on boot argument

Submitted upstream to lkml by Dmitry

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Factory Images

Now, installing the latest AOSP master build is easy!

https://developers.google.com/android/nexus/images-preview#hikey

```
# Set the J15 3-4 jumper
$ unzip <filename>
$ cd <filename>
$ ./flashall.sh /dev/ttyUSBn
# Unset J15 3-4 jumper & reboot
```

That's it!



Common.git AOSP Efforts

Appended dtb support (Image-dtb)

Integrated upstream timerslack_ns support

Align with upstream cgroup migration update

-stable merged android-4.4.y branches

EAS forward ported from android-3.18 to android-4.4

Deep review of common/android-4.4 tree

Sending reverts for obsolete features

Prep-work for next LTS android-4.9 tree

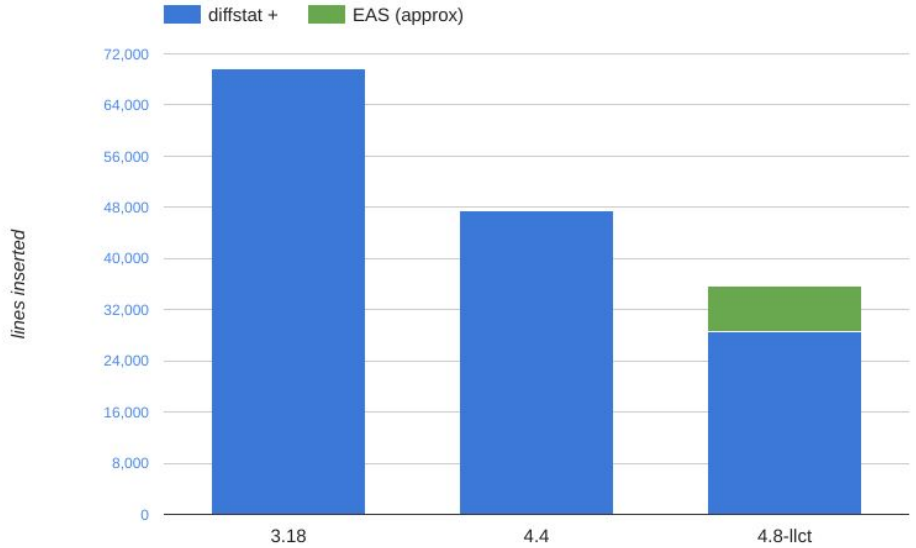
Android upstreaming...



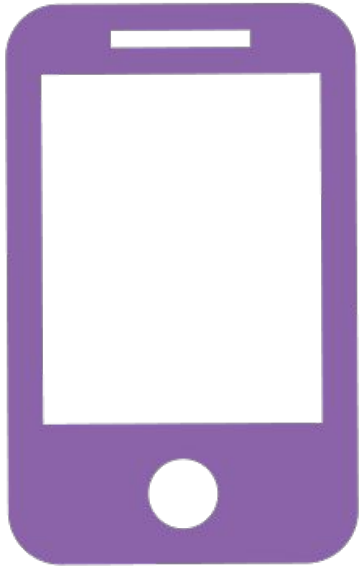
Amit Pundir's talk on state of the Android patches:

Video: https://www.youtube.com/watch?v=lX14Y6lG4Aw

Slides: https://goo.gl/XUzXkm



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LMG
Mobile

Generic Build Integration

Not AOSP, but HiKey is also used in Rob Herring's generic build project, which supports multiple devices (and architectures) out of a single build directory using Kconfig

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TODOs

- Moving forward to next android-4.9 LTS kernel
- Work with Mali devs to avoid custom tweaks for Hikey
- Op-tee & Trusty coexisting in AOSP
- A/B updates partitioning & switching
- Memory reductions to help 1GB variant
- Fixing bugs as they crop up.

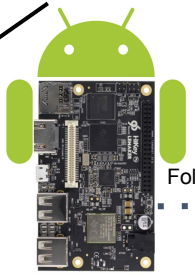
Why this is useful...



Latest flagship devices (Mar 2016)



2015 Nexus devices (Oct 2015)



HiKey in AOSP (Apr 2016)

Following upstream kernels

2 years, 4 months

1 year , 3 months

4 months

v3.10 Released
June 2013

v3.18 Released
Dec 2014

v4.4 Released
Jan 2016

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Regressions fixed in android-4.4

- Two separate xt_qtaguid bugs
- USB eth adapter regression
- USB Configfs gadget fixes
- PTP null pointer deref fix
- Missing cpuset allow_attach hooks
- Performance regression in cgroup migration
- UID routing enum collision



Regressions found and fixed upstream

- /sys/module/mmcblk path ABI change
- wlcore_op_get_expected_throughput null ptr dereference
- Dwc2 gadget TX FIFO breakage
- Missing CAP_WAKE_ALARM in AOSP userspace
- fib_rules routing collision w/ upstream
- iptables alignment breakage
- Cgroup migration permissions issue

HiKey Upstreaming Status

4.6:

- PMIC
- thermal

4.7:

- eMMC
- uSD
- USB
- Wifi
- DRM display

4.8:

- power-key
- RTC
- media-reset
- adv7533

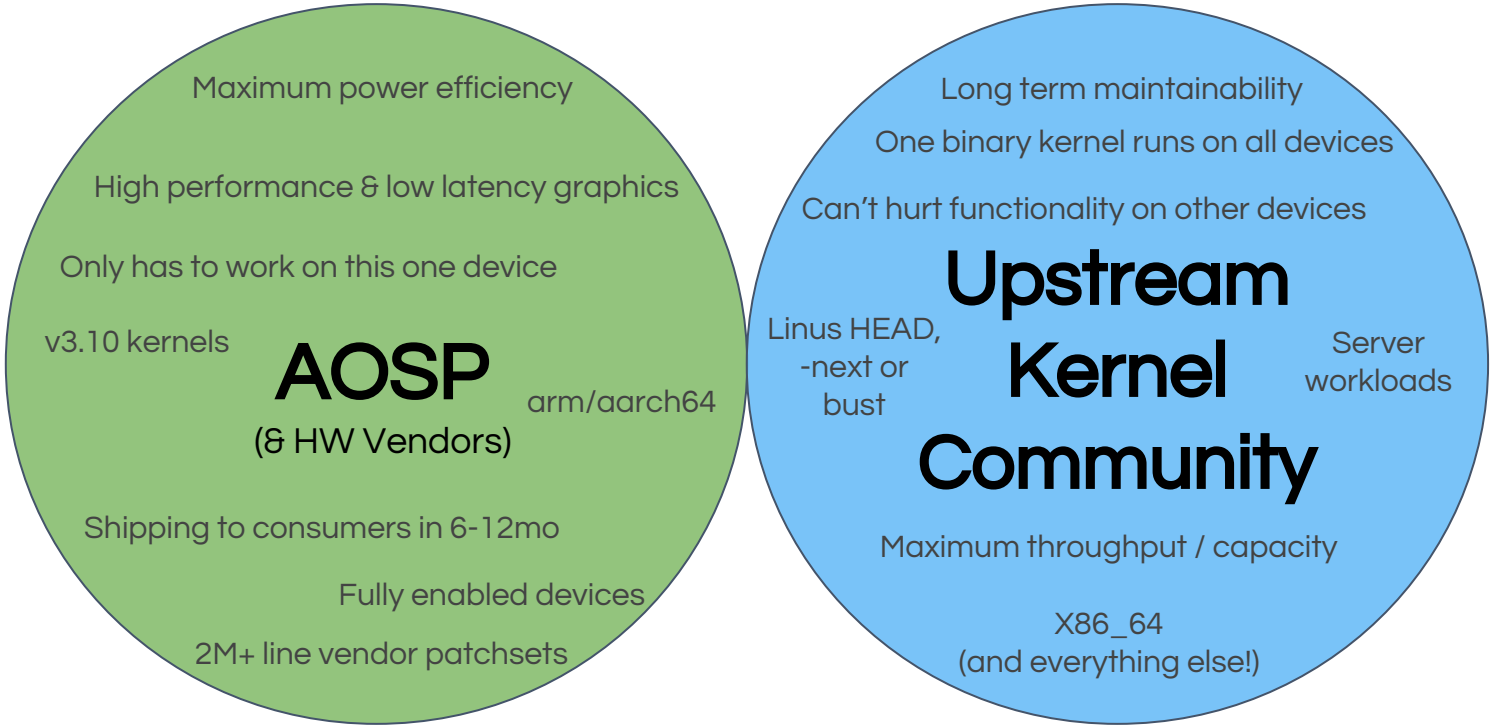
4.9:

- K3DMA fixes
- HDMI dts
- Reboot-reason dts,
- Pstore dts
- Kconfig fix

HiKey Upstreaming TODOs (~35 patches)

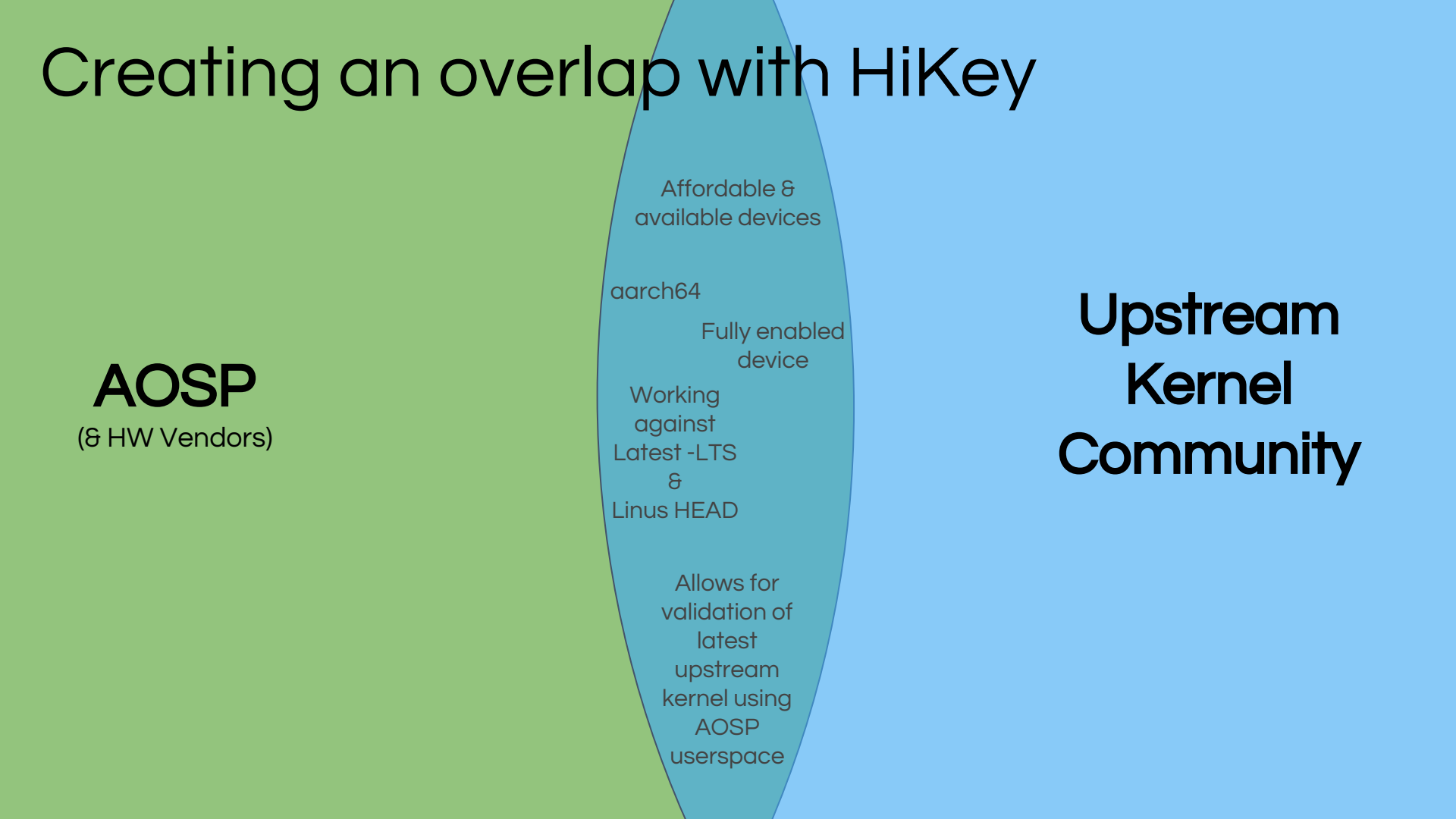
HDMI audio support	[5 patches]
USB speed autonegotiation / fixes	[5 patches]
Adv7511 improvements	[4 patches]
Android uid-routing workarounds	[2 patches]
Upstream Bluetooth solution for AOSP	[4 patches]
~~Mali driver~~ (we can dream!)	[~15 patches]

Differing focuses



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Creating an overlap with HiKey





Thanks!

Questions?
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